Allan Suraj A

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Education

PES University **Bengaluru, India**
B.Tech in Electronics and Communication Engineering | CGPA: 8.08 / 10 *2023 – 2027 (Expected)*
- **Relevant Coursework:** Machine Learning, Database Management Systems, Computer Architecture, Digital System Design

Technical Skills

Languages: Python, C, C++, SQL, MATLAB
AI / Machine Learning: Deep learning (CNNs, YOLOv8, ResNet50), computer vision, object detection, RAG pipelines, local LLM deployment (Ollama), Grad-CAM
Software Development: React, Git, Linux, relational databases, debugging & profiling
Hardware & Systems: SystemVerilog/Verilog RTL, Xilinx Zynq SoC, Vivado, FPGA prototyping, Cadence Virtuoso
Security: OS hardening, security-policy implementation, attack-surface analysis, configuration management

Experience

Ivory Solutions (Remote) **Bangalore, India**
Cybersecurity Intern *Jul 2026 – Sep 2026*
- Worked on IvoryHunt, a job-seeking platform the company was building.
- Set the rules for how candidate and recruiter data is stored and who is allowed to reach it.
- Advised on the product's security and storage decisions before anything was built.

Bharat Electronics Limited (BEL) **Bengaluru, India**
Student Intern — Network Systems & Cybersecurity *Jul 2025 – Aug 2025*
- Hardened Windows 10 machines in a defence software environment by checking configurations against security baselines and applying group policy controls.
- Worked alongside senior engineers on OS-level performance tuning and day-to-day system administration in an enterprise setup.

PESU Venture Labs **Bengaluru, India**
Research Intern — Deep-Tech Venture Research *Jun 2025 – Aug 2025*
- Researched semiconductor and deep-tech trends to put together problem statements that could support a startup.
- Spoke with senior people in the semiconductor industry to check whether those assumptions held up, then wrote them into structured briefs.

Projects

PRISM — Offline Edge AI Assistant for Aerospace MRO | *Python, Flask, RAG, YOLOv8* **Tata InnoVent-27**
- Built an offline diagnostic assistant for aircraft maintenance technicians that answers a symptom query from 4,769 passages indexed out of FAA advisory circulars, shows the passages it used, and refuses to answer when the manuals do not cover it.
- Ran Phi-3 Mini locally through Ollama so nothing leaves the machine, and added telemetry-driven risk scoring, a fine-tuned YOLOv8n defect detector, and a case store keeping resolved diagnoses for similarity search. Flask backend, React frontend, SSE streaming.

Uncertainty-Aware Medical Abnormality Detection & Grounding | *Python, Deep Learning, Computer Vision*
- Trained a model to find and mark chest X-ray abnormalities straight from the image, without needing an earlier clinical report to work from.
- Scored each prediction for confidence and passed the uncertain ones to a human instead of answering, with Grad-CAM so a radiologist can see what the model looked at.

Hardware-Software Co-Design for CNN Acceleration | *Xilinx Zynq SoC, Winograd* **Jun 2025 – Present**
- Porting a CNN inference pipeline onto a Xilinx Zynq SoC, swapping direct convolution for a modified Winograd algorithm to bring down 2D convolution latency on edge hardware.
- Splitting the work between the programmable logic and the on-chip processor, and laying out the datapath and memory hierarchy to fit the available logic and BRAM.

EO/IR Smart Surveillance System | *Python, YOLOv8, ResNet50* **HAL Aerothon 2025**
- Ran threat detection across RGB, thermal and aerial imagery using four models, picking the one that suits the input instead of pushing everything through a single detector.

Hackathons

Terrathon — FinSync | *React, Accessible UI* **Oct 2025**
- Built a banking app in React for senior citizens, with a high-contrast interface and Kannada support. Payments above a set amount stay on hold until a trusted contact approves them.

Horcrux — SuperPowerGlasses | *Computer Vision, Embedded Systems* **Sep 2024**
- Put together smart glasses that send camera and mic input to a laptop and return audio and visual prompts, aimed at users with autism or visual impairment.

Certifications

Google Cybersecurity Professional • ML Specialization (Coursera) • Semiconductor Manufacturing, IISc